HW7

a)

2007 Artist 1: Planet [HD 189733b]

* Hot-Jupiter, tidally locked, dark atmo, stellar-flux transmitted

- Abstract: 'Hot-Jupiter' planets expected to be tidally locked bc they are so close (<0.05 AU) to their parent star, resulting in permanent daysides and nightsides.

- Main neighbor was monitored over ~33 Hr period using a 8Mm channel of Infared array camera, with cadence of 0.4 s. observations spanned little over half planets orbit, starting 2.6 Hrs before start of trans (when planet moves in front of star) & ending 1.9 Hrs after secondary eclipse (planet behind star). 278,528 32x32 Imgs collected. gradual, illumination-dependent, detector induced rise of up to 10% in signal measured in individual pixels over time. Illumination heavy pixels converged to constant value within 2 Hrs of OBS, lower flux pixels approaching linearly.

data analysis: peak hemisphere-integrated brightness before opposition (directly opposite to host star) corresponding to eastern shifted hotspot (from substellar pt.). Secondary eclipse happens later than expected (eliminates circular orbit). Results argue for dark atmospheres at visible wavelengths, which absorb a lot of stellar flux, and also able to transport this energy to the darkside.

b)

2008 Artist 2: Broadband Infared Emission Spectrum of [redacted]

Abstract: examined exoplanet neighbor in infrared over 2 secondary eclipses. Using observations able to estimate 5 bands of relative eclipse depth, and also then find the 5 corresponding planet-to-star flux ratios as well. Exoplanet neighbor does not behave like a blackbody, enforced by opacity window near 4Mm as predicted in models. Unlike other exoplanet (209458b), exoplanet neighbor does not match models that include a temp. inversion. Provides (very tentative) support that Hot-Jupiters fall into 2-classes, in which a thermal inversion layer is present for more strongly irradiated objects. M-dwarf companion nearby (noted possible error) conclude (both models: flux confined to dayside vs. distributed) shows clear deviation from planck curve. especially greatest at 3.6 Mm, arising from opacity window at 4 Mm. many different Hot-Jupiter exoplanets (diverse). Thermal inversion might be caused by chem. compounds in Atmosphere (TiO, VO), further study warranted.

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Other studies (pre-James webb)

* Not a blackbody, no thermal inversion, using IR obs

# Mon Recent

c)

2013

Article 3: Albedo measurements with Hubble imaging spectrog. of Vis. λ

The Deep Blue color of exoplanet □

-Abstract: Secondary eclipse observation of exoplanet □ across 290-570nm using spectrograph. Measured Geometric Albedo at 1σ confidence, which decrease at longer wavelengths when using 6 λ bins over same range. Can be interpreted as optically thick clouds over dayside hemisphere (with sodium absorption suppressing scattered light beyond ~450nm. Best fit albedo values suggest □ is blue at visible wavelengths

Main notes: Geometric Albedos are ratio of a celestial bodys actual brightness seen from the light source to the brightness of idealized (flat, fully reflecting, diffusively scattering) disk of same size. role of study was to gauge role of clouds/haze in □ atmosphere. Clouds/haze in atmosphere has been established possible cause of high Albedo. Measured Reflection signal by monitoring change in brightness of star-Planet system during second eclipse. secondary eclipse signal directly related to dayside Hemisphere brightness. □ prev. measured at IR λ (Article 1), this new obs. is the first at short λ, where thermal emissions negligible, so any detected light is due to scattering by atmosphere. Variations in star brightness (instead of planetary eclipse) could be responsible for measured signal especially pertinent of □ which is an active M-dwarf, however, flux variations can be estimated using new index of -2.3 obtained by McQuillan. Seems plausible that reflection signal caused by same scattering species that produces Rayleigh profile in transmission spectrum. Given the strong atmo. circulation expected for hot-Jupiters. Study demonstrates potential of secondary eclipse obs. with HST/STIS (Hubble/Spectrograph), study adds to understanding of of □ atmo. & constraints

2024

d) Article 4: Retrieves on NIRCam transmission and emission spectra of □ with PLATON 6, GPU code for JWST era

-abstract: presents 2.4-5.0 μm JWST/NIRCam (near-IR cam) emission spectrum of □, also talking about prev. published transmission spectrum of same λ. Comparing two transmissions, found they were consistent across both geometries (the methodology and C/O estimates). Possible thermal inversion at low pressures on dayside.

-body: New code version of PLATON for JWST at specific λ for the transmission and emission spectrum indicate a metallicity (M/H) 2.6-6.8× solar and C/O ratio of 0.42, the C/O spectrum being 3σ consistent (transmission), whereas the emission spectrum was much narrower and inconsistent. Methane found to be depleted at terminal and thermal inversion tentatively indicated on dayside (1.8σ). Possible complications include atmosphere (clouds), deficiencies for M/H and C/O as well as opacity resolution.